MIL-C-915E
AMENDMENT-2
30 May 1980
SUPERSEDING
AMENDMENT-1
5 April 1973

MILITARY SPECIFICATION

CABLE AND CORD ELECTRICAL, FOR SHIPBOARD USE

GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-C-915E, dated 1 August 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

2.1: Add the following Military Specification:

"MIL-I-45208 - Inspection System Requirements."

PAGES 3 and 4

* 2.2: Add the following publication under American Society for Testing and Materials:

"ASTM D 2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer."

* 2.2: Add the following:

"INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS
"IEEE-STD-383 - IEEE STD for Type Test of Class 1E Electric Cables, Field Splices and Connections for Nuclear Power Generating Stations.

"(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, Inc., 345 East 47th Street, New York, NY 10017.)"

PAGE 5

* "3.3.3.1, last sentence: Delete.

PAGE 6

3.3.9, first sentence: Delete and substitute: "Tie cords shall be of cotton or synthetic fiber having a maximum diameter of 0.065 inch and a minimum breaking strength of 30 pounds."

FSC 6145

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3.4.1: Delete and substitute:

"3.4.1 Conductor stranding. The size and quantity of individual conductor strands and the total circular-mil area of the conductor, unless otherwise specified on the specification sheet, shall be in accordance with tables I, II, or III, as applicable, in accordance with the conductor size designation indicated on the specification sheet. When AWG sizes are specified on the specification sheets for conductors, they shall be in accordance with ASTM B 8."

PAGE 8

Table III, column 2: Delete all values under "Rope" and substitute the following:

Number of strands Minimum
" 49
228
49
304
336
209
49
532
304
133
2107
418
2646
259
3325
259
648
760
427
988
259
1254
6384
259
2052
259
427
427
4033"

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Table IV: Add the following as color, conductor or group No. "50".

Color, conductor or group No.	Background or base color	First tracer color	Second tracer color
"50	Black	Orange	Red"

PAGE 12

* Add the following paragraphs:

* "3.4.9.1.5 Twisted pair identification code. This code shall consist of numbers in sequence running from 1 through the number corresponding to the total quantity of twisted pairs in the cable. Both conductors in each pair shall be numbered the same, denoting the sequence number of the pair. Distinction between the two conductors is provided by different colored insulation. Conductors of a cable with a single pair need not be numbered.

* "3.4.9.1.6 Twisted triad identification code. This code shall consist of numbers in sequence running from 1 through the number corresponding to the total quantity of twisted triads in the cable. All three conductors shall be numbered the same, denoting the sequence number of the triad; distinction between the three conductors is provided by different colored insulation. Conductors of a cable with a single triad need not be numbered."

PAGE 13

* Add the following paragraph:

"3.4.9.2.6 Method 6. Identification method 6 shall consist of numerals printed in ink on the conductor insulation. For conductors having a jacket directly over the insulation, the numerals may be printed in ink on the jacket, at the manufacturer's option. White ink shall be used for a red or black background; black ink shall be used for a white background. Numerals shall be upright on the logitudinal axis of the conductor (see figure 6). Numeral width shall be proportional to conductor, outside diameter (o.d.) as follows:

Conductor (o.d.) (inch)	Numeral width (inch)
0.040 - 0.050	0.020
.051 - .070	.025
.071 - .095	.030

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Numeral height shall be 2-1/2 to 3 times numeral width. Each numeric legend shall be underlined. Two-digit legends shall have the bottom numeral underlined. Legends shall be alternately inverted and shall be repeated at intervals not greater than 1-1/2 inch.

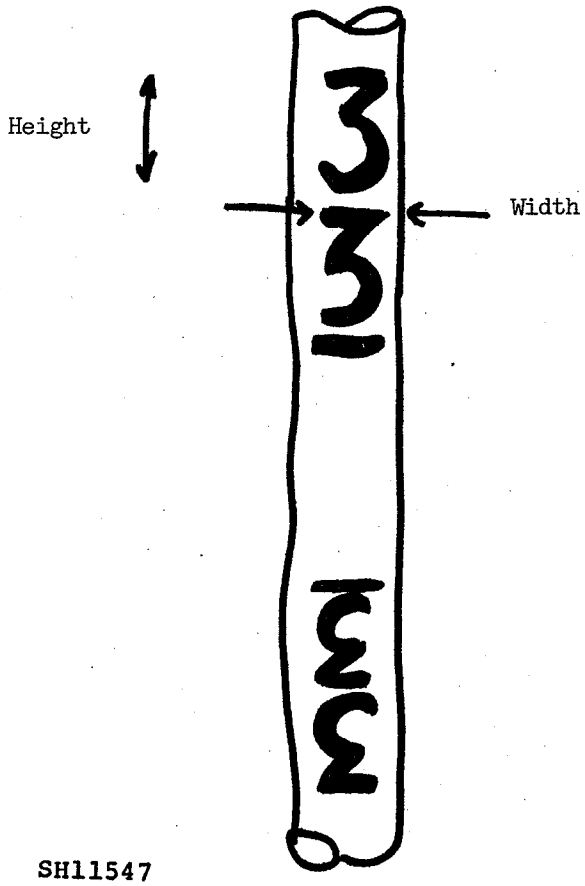


FIGURE 6. Conductor of pair number 33."

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Add the following paragraph as 4.1.1:

"4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208."

PAGE 17

* Table VII: Delete and substitute:

* "Table VII - Qualification specimens and groups.

Group No.	Qualification		Types and Sizes Comprising the Group
	Flame Propagation 1/	Type and Size 2/	
1		DLT	DLT
2		DRW	SRW, DRW, TRW
3		DSS-3	All sizes of DSS, TSS, FSS, 5SS and 7SS
4		DSWS	DSWS
5		ECM	ECM, MSP and MSPW
6		JAS	JAS
7		MCOS-6 . . .	All sizes of MCOS
8		MCSF.	MCSF
9	MDU-6 & MDU-23	MDU-23 . . .	All sizes of MDU
10		MHOF-30 . . .	All sizes of MHOF
11		MMOP.	All sizes of DCOP, TCOP and MMOP
12		MNW-10 . . .	All sizes of MNW
13	MSCU-10, MSCU-44 & ~~TSGU-14~~	MSCU-10 . . .	All those sizes of DSGU, TSGU, FSGU, MSCU, 7SGU, TCJU and TCTU which have conductor sizes 23 and smaller
14		MWF-24 . . .	All sizes of MWF
15	PBTMU-15	PBTMU-15 . .	All sizes of PBTMU and TSP
16	PI-12, TPS-9	PI-12	All sizes of PI, DPS, TPS, FPS and 7PS
17		S2S	S2S
18	TCTX-7	TCTX-7 . . .	All sizes of TCJX, TCKX and TCTX
19		THOF-9 . . .	Sizes 42 and smaller of types DHOF, THOF and FHOF, except type-and-size THOF-14
20		THOF-150 . .	Sizes 60 and larger of types DHOF, THOF and FHOF; type-and-size THOF-14; type CVSF

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Table VII - Qualification specimens and groups. - (Continued)

Group No.	Qualification		Types and Sizes
	Flame propagation 1/	Type and size 2/	Comprising the group
21		TNW-150	All sizes of DNW, TNW, FNW, 4NW8 and 8NW6
22	TPNW-20	TPNW-20 . . .	All sizes of TPNW
23		TPU	TPU
24	TSGU-50 & TSGU-150	TSGU-150 . .	All those sizes of SSGU, DSGU, TSGU, FSGU and 6SGU which have conductor sizes 50 and larger
25		TTOP-10 . . .	All sizes of TTOP
26		TTRS-8 . . .	All sizes of TTRS
27	TTSU-10	TTSU-10 . . .	All sizes of TTSU
28	MS & MU	MS	All sizes of MS, MU, 2SJ, 3SJ and 4SJ
29		1S50MU-16 . .	All sizes of 1SAU, 1SMU, 1S50MU, 1S75MU and 1SU.
30		2AU-40 . . .	All sizes of 2AU, 2U, 2SU 3SU and 3U.
31		2SWAU-10 . .	All sizes of 1SMWU, 1SWU, 2SWAU, 2SWU, 2WAU, and 3SWU.
32		2SWF-7 . . .	All sizes of 1SWF, 2SWF and 3SF.
33	5KVTSGU-100	5KVTSGU-100 .	All sizes of 5KVTSGU
34		2UW-42 . . .	2UW
35		TPUM	TPUM

1/ Types and sizes required for flame propagation test for group qualification.

2/ Type and size required for group qualification. Additional type and size requirements for flame propagation test as indicated. Suggested specimen length is 200 feet for all qualifying types and sizes, except 2SWAU-10, for which it is 500 feet. Additional length of cable required where flame propagation test is specified."

PAGE 20

4.6.1: Delete and substitute:

"4.6.1 Sampling procedure for comparison inspection. Once a manufacturer's product has been qualified, the manufacturer shall submit one specimen for comparison inspection from the first lot of each type and size that he produces. Thenceforth, the manufacturer shall submit one specimen of each type and size that he produces within each calendar year. The entire lot of cable or cord which is represented by the specimen submitted for comparison inspection shall be retained by the manufacturer until he receives, from the Government, notice of the results of comparison inspection.

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4.6.1.2: Add the following items (a) through (g):

"(a) Manufacturer's name.
"(b) Cable type-and-size designation.
"(c) Applicable specification (that is, the MIL-C-915 specification sheet number).
"(d) The test or qualification reference.
"(e) Footage represented by the specimen.
"(f) The number of the coil or reel from which the specimen was taken.
"(g) Either the manufacturing number or the order number assigned by the customer."

PAGES 25 and 26

4.8.8 through 4.8.8.3.1: Delete and substitute:

"4.8.8 Cable aging and compatibility (125°C). This test is applicable to cables having either thermosetting insulation or silicone rubber-glass tape insulation, and is for the purpose of detecting any significant degradation in the characteristics of the cable or its components, resulting from either component incompatibility or prolonged overheating. The cable aging may be accomplished either by the current overload method or by the hot air oven method.

"4.8.8.1 Specimens. The specimen for the current overload method shall consist of a length of completed cable approximately 40 feet long. The specimens for the hot air oven method shall consist of two lengths of completed cable: One length approximately 30 feet long and one length 2 feet plus 24 times the overall diameter of the specimen.

"4.8.8.2 Apparatus. The test apparatus for both methods shall consist of a heat chamber or oven, a means of measuring and controlling the chamber or oven temperature within plus or minus 3°C of the specified temperature for a continuous period of 400 hours, an internal mercury thermometer and a rack or other means of supporting the specimen or specimens within the chamber or oven. For the current overload method there shall be a source of current capable of supplying the required amount of current for a continous period of 400 hours. For both methods there shall also be a mandrel having a diameter of approximately 12 times the specimen diameter.

"4.8.8.3 Procedure. The procedure for each method shall consist of two parts: (1) heat aging; (2) bending.

"4.8.8.3.1 Part one - heat aging. (May be accomplished by either of the following methods).

(a) Current overload method. Prior to heat aging, the specimen shall be given the voltage withstand and the insulation resistance tests. The maximum values of insulation resistance measured for each conductor shall be recorded. During aging, conductor temperature shall be determined either by direct measurement or by the measurement of change of copper resistance. For the direct measurement method, a thermocouple shall be inserted through a small

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knife puncture in the cable, approximately midway between the ends of the specimen. For cables having only one layer of conductors, the thermocouple shall be placed in contact with the strands of any conductor. For cables having more than one layer of conductors, the thermocouple shall be placed in contact with the strands of any conductor in the innermost layer. Thermocouples shall also be attached to the jacket of unarmored cables. In the case of armored cables, the thermocouples shall be attached to the armor. The specimen shall be placed into the chamber so as to have a straight section of approximately 7 feet, and so as to permit free circulation of convection air currents. The chamber temperature shall be maintained at 50°C ± 3°C during the 400-hour period. The amplitudes of loading currents shall be such as to maintain each conductor at a temperature of 125°C ± 5°C continuously for 400 hours. Readings of time, chamber temperature, jacket or armor temperature, and the current in each conductor shall be recorded at 15-minute intervals during the first hour, at 1-hour intervals during the next 5 hours, and twice daily thereafter for the duration of the test. Record shall also be made of any gas or fumes emanating from the specimen, and of any oozing of material from the ends of the specimen. In the case of armored cable, record shall also be made of any exudation of the jacket material through the armor. At the end of the 400-hour test period, the specimen shall be allowed to cool to room temperature. Then it shall be given the voltage withstand and the insulation resistance tests. The maximum insulation resistance values measured for each conductor shall again be recorded. From the portion of the 40-foot specimen which was kept straight during the heat aging, a specimen having an approximate length of 2 feet plus 24 times the overall diameter of the specimen shall be taken. This specimen shall be used for the bending procedure. From the remainder of the 40-foot specimen, an additional specimen shall be taken an given the cable filler removability test.

(b) Hot air oven method. Prior to heat aging, the 30-foot specimen shall be given the voltage withstand and the insulation resistance tests. The maximum value of insulation resistance measured for each conductor shall be recorded. The two specimens shall be suspended horizontally in the oven, with the shorter specimen kept straight. The oven temperature shall be maintained at 125°C ± 3°C during the 400-hour period. Record shall be made of any gases or fumes emanating from the specimens and of any oozing of material from the ends of either specimen. In the case of armored cable, record shall also be made of any exudation of jacket material through the armor. At the end of the 400-hour test period, the specimens shall be allowed to cool to room temperature. The shorter specimen shall be used for the bending procedure. The 30-foot specimen shall again be given the voltage withstand and the insulation resistance tests. The maximum value of insulation

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resistance measured for each conductor shall again be recorded. One additional specimen shall then be taken from the 30-foot specimen, and given the cable filler removability test."

4.8.8.4, items (b), (c), and (d), line 1: Delete "40-foot".

PAGE 29

4.8.16.2.4, line 3: Delete "certerlines" and substitute "centerlines".

PAGES 34 and 35

4.8.22.3.2, fourth sentence: Delete and substitute: "Each specimen shall be stretched at a uniform rate of 20 inches per minute (unless otherwise specified on the specification sheet), until rupture."

PAGE 37

* Add the following paragraphs 4.8.31 through 4.8.31.3 and 4.8.32 through 4.8.32.2:

* "4.8.31 Flame propagation. The cable shall be tested in accordance with IEEE-STD-383 flame test procedure using the ribbon gas burner, vertical tray configuration.

* "4.8.31.1 Qualification specimens. The cable type and size required for group qualification shall be in accordance with table VII.

* "4.8.31.2 Observation. Cables which propagate the flame and burn the total height of the tray above the flame source fail the test. Cables which self-extinguish when the flame source is removed pass the test. Cables which continue to burn after the flame source is removed shall be allowed to burn in order to determine the extent.

* "4.8.31.3 Report. The report shall include the cable type, size, flame temperature, length of time of afterburn, overall distance of damage to the jacket above the level of the burner, and overall distance of damage to the individual conductors, above the level of the burner, dripping or flaming material falling from the cable onto the floor.

* "4.8.32 Durometer hardness (type A). The durometer hardness of the material shall be tested in accordance with ASTM D 2240, type A.

* "4.8.32.1 Specimen preparation. The specimen shall be a single thickness of material conforming to the requirements of ASTM D 2240. Preparation of the specimen shall effectively represent the curing cycle of the material as used in manufacture of the cable.

* "4.8.32.2 Report. The report shall be in accordance with ASTM D 2240 and shall include details regarding sample preparation and cure cycle."

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PAGE 41

4.9.5.1, line 5: After "all other conductors", add "and".

PAGE 44

4.9.8.2, first sentence: Delete and substitute: "The test voltages shall be as specified in the applicable specification sheet. The point of application shall be conductor to conductor unless otherwise specified on the applicable specification sheet."

PAGE 52

* Index, after "drip" add the following listing:

	Paragraph	Page
"Flame propagation	4.8.31	37"

NOTE: The margins of this amendment are marked "*" to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last prevous amendment.

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Navy - SH
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